Lesson plan · for the teacher

Bits and bytes

IoT · Level 2 · Lesson 3

Time
40 min

Materials
Paper or card for five dot cards per group; Markers or pens; The board, for a place-value chart

Learning goals

- Count in binary with place values, from 0 to 255.
- Explain why more bits give finer readings (10 bits give 1,024 levels).
- Estimate how much data a reading, a message and a photo take.

Activity: Binary dot cards

1. Warm-up (5 min): Show (or describe) a long Arabic SMS split into three parts and the same text in English in one. Ask: why would the phone company count them differently?
2. Teach (8 min): Draw five boxes labelled 16, 8, 4, 2, 1. Light up boxes to show 5 (101) and 6 (110). Explain that 8 bits make a byte, 0 to 255.
3. Cards (12 min): Each group draws 1, 2, 4, 8 and 16 dots on five cards. Call out numbers (9, 20, 31); groups show them face up for 1 and face down for 0. The first correct group scores.
4. Apply (10 min): Pupils write their age in binary, swap with a partner and check by adding the dots.
5. Wrap-up (5 min): Ask: why can't one byte hold 300? (The most 8 bits can hold is 255.) Exit ticket: 13 in binary.

Make one set of cards to show the class. The dots must be clear from the back of the room: 16 dots fit in a 4 × 4 grid.

Discussion

- Where do your pupils meet bits and bytes in daily life?
- Why do data bundles count megabytes, not messages?
- What would a sensor need more bits for?

Exit ticket

- Write 13 in binary, and say why 8 bits stop at 255.

Try it at school tomorrow
Binary dot cards: Make five cards per group with 1, 2, 4, 8 and 16 dots. Call out numbers and let groups race to show them, face up for 1 and face down for 0. Then each pupil writes their age in binary, and the class works out why 8 bits can't hold more than 255.

Practice questions · for you

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1 **A weather station sends one temperature reading every 10 minutes over mobile data. Roughly how much data is that?**

◯ Several megabytes a day ◯ Kilobytes a day: tiny ◯ About one photo's worth every hour

2 **In binary, 1000 is a bigger number than 111.**

◯ True ◯ False

3 **Complete the sentence about one byte.**

With 8 bits you can make _____ different values, from 0 to _____.

Word bank: 8 · 128 · 256 · 255

4 **Match each number of bits to the range of values it can hold.**

1 bit	• • • •	0 to 1023
4 bits	• • • •	0 to 4095
8 bits	• • • •	0 to 15
10 bits (Arduino Uno)	• • • •	0 to 1
12 bits (ESP32)	• • • •	0 to 255

5 **Put these in order of size, smallest first.**

☐ One SMS (up to 140 bytes)

☐ A phone photo (about 3,000,000 bytes)

☐ One temperature reading (about 2 bytes)

☐ A page of a book (about 2,000 bytes)

✎ **Which data costs your school or your pupils' families the most: messages, photos or videos? Where could a number replace a picture?**

Check your answers

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1 **Kilobytes a day: tiny**

That's 144 readings a day at a few bytes each.

2 **True**

1000 is 8; 111 is 4 + 2 + 1 = 7. The first 1 sits in a higher place than any of the three, so it's bigger, even though it has fewer 1s.

3 **With 8 bits you can make 256 different values, from 0 to 255.**

Each bit doubles the count: $2^8 = 256$ values.

4 **1 bit — 0 to 1**
4 bits — 0 to 15
8 bits — 0 to 255
10 bits (Arduino Uno) — 0 to 1023
12 bits (ESP32) — 0 to 4095

n bits hold 2^n values, from 0 to $2^n - 1$.

5 **Top to bottom: 2, 4, 1, 3**

Each one is more than ten times bigger than the one before, and the photo is over a thousand times bigger than the page.